

ROVANIEMI AP, Finland

WMO: 028450

Lat: 66.562N

Lon: 25.822E

Elev: 194

StdP: 99.01

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.9	(c) -23.1	(d) -28.5	(e) 0.3	(f) -25.8	(g) -25.4	(h) 0.4	(i) -23.0	(j) 11.2	(k) -4.4	(l) 10.1	(m) -4.8	(n) 2.6	(o) 70	(p) 0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.7	(c) 24.7	(d) 16.3	(e) 22.8	(f) 15.3	(g) 21.0	(h) 14.5	(i) 18.0	(j) 22.0	(k) 16.8	(l) 20.8	(m) 15.6	(n) 19.2	(o) 4.1	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	12.0	19.6	15.2	11.0	18.1	14.1	10.2	17.1	51.7	21.9	47.8	20.8	44.4	19.1	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.5	(b) 8.3	(c) 7.4	(e) -27.9	(f) 27.1	(g) 3.8	(h) 2.1	(i) -30.6	(j) 28.6	(k) -32.8	(l) 29.8	(m) -34.9	(n) 31.0	(o) -37.7	(p) 32.5		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-27.9	19.6	3.7	1.5	-30.6	20.7	-32.8	21.6	-34.9	22.4	-37.6	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-10.7	-10.3	-5.7	0.4	6.4	12.3	15.8	13.2	8.0	1.1	-4.6	-8.1
	DBStd	10.29	6.69	6.43	4.61	3.62	4.58	3.90	3.36	3.29	3.49	4.22	5.12	6.34
	HDD10.0	3504	643	568	487	288	132	20	1	8	79	277	439	562
	HDD18.3	6151	902	801	746	538	370	184	93	162	311	535	689	820
	CDD10.0	416	0	0	0	0	20	89	180	108	18	0	0	0
	CDD18.3	21	0	0	0	0	1	3	14	3	0	0	0	0
	CDH23.3	122	0	0	0	0	4	24	78	16	0	0	0	0
	CDH26.7	13	0	0	0	0	0	3	8	1	0	0	0	0
(14) Wind	WSAvg	3.7	3.7	3.9	4.0	3.7	3.8	3.6	3.4	3.3	3.5	3.6	3.7	3.8
(15) Precipitation	PrecAvg	535	37	30	30	27	33	51	61	70	60	54	45	36
	PrecMax	723	93	73	73	85	90	115	133	139	123	108	87	58
	PrecMin	394	9	11	2	6	5	8	12	18	7	25	13	8
	PrecStd	84	19	17	17	17	17	29	32	30	28	23	16	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	2.7	6.0	13.3	23.6	25.9	28.0	25.3	18.2	10.7	6.1	3.1
		MCWB	1.0	1.5	2.6	7.1	13.2	16.3	18.6	17.9	13.5	9.4	5.6	2.6
	2%	DB	0.2	0.7	3.4	10.1	20.1	23.0	25.5	22.7	16.1	9.4	4.2	1.6
		MCWB	0.0	0.2	1.0	5.2	12.1	14.3	16.9	15.8	12.2	8.5	3.8	1.3
	5%	DB	-0.8	-0.3	1.7	7.8	16.6	21.0	23.6	20.6	14.6	8.0	2.7	0.5
		MCWB	-1.0	-0.5	-0.1	3.9	10.1	13.3	16.2	15.0	11.7	7.2	2.5	0.3
	10%	DB	-2.1	-1.6	0.2	5.8	14.0	19.0	21.7	18.7	13.0	6.7	1.4	-0.3
		MCWB	-2.2	-1.9	-0.8	2.5	8.6	12.3	15.4	14.3	11.1	6.1	1.2	-0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.7	2.8	7.9	14.4	18.1	20.1	19.1	14.6	10.1	5.8	2.7
		MCDB	1.4	2.3	5.1	12.5	20.7	23.1	24.5	23.7	16.7	10.4	5.9	2.9
	2%	WB	0.0	0.2	1.3	5.8	12.6	16.1	18.6	17.4	13.2	8.8	3.9	1.2
		MCDB	0.2	0.4	2.8	8.9	19.1	21.0	22.7	20.7	15.1	9.3	4.2	1.4
	5%	WB	-0.9	-0.7	0.2	4.2	10.7	14.5	17.4	16.0	12.3	7.2	2.5	0.2
		MCDB	-0.8	-0.5	1.1	7.2	15.7	18.8	21.6	18.7	13.9	7.8	2.7	0.4
	10%	WB	-2.2	-1.9	-0.5	2.8	9.1	13.3	16.3	14.9	11.3	6.1	1.2	-0.4
		MCDB	-2.1	-1.6	0.2	5.2	13.3	17.4	20.3	17.8	12.7	6.6	1.3	-0.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.5	6.9	7.1	7.8	7.6	7.7	7.2	6.1	4.2	4.5	5.3
		MCDBR	5.4	4.8	7.5	9.5	12.0	10.6	10.2	9.7	8.1	4.9	4.0	4.6
		MCWBR	5.4	4.5	5.6	5.9	6.1	4.9	4.3	4.4	4.9	4.1	3.9	4.5
	5% WB	MCDBR	5.5	4.5	6.1	8.9	11.0	8.9	9.0	8.0	6.7	4.4	3.9	4.5
		MCWBR	5.5	4.4	4.9	5.7	6.1	4.9	4.7	4.7	4.6	4.0	3.9	4.5
(40) Clear-Sky Solar Irradiance	taub		0.193	0.261	0.293	0.327	0.338	0.340	0.354	0.337	0.313	0.285	0.346	N/A
	taud		1.932	2.110	2.178	2.252	2.456	2.481	2.459	2.488	2.540	2.353	2.677	N/A
	Ebn at Noon		207	578	757	829	864	870	843	822	762	567	221	0
	Edh at Noon		15	53	83	101	93	93	92	81	61	42	16	0
(44) All-Sky Solar Radiation	RadAvg		0.07	0.55	1.98	3.67	4.72	5.18	4.85	3.53	1.85	0.64	0.13	0.00
	RadStd		0.01	0.10	0.20	0.32	0.43	0.39	0.46	0.32	0.20	0.13	0.02	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page